PRIME FOCUS SET-UP
This technique uses an adapter to place the camera in the eyepiece position, with or without the eyepiece present. A motor-driven equatorial mount is needed to keep the target object in the field of view.

PRIME-FOCUS ASTROPHOTOGRAPHY

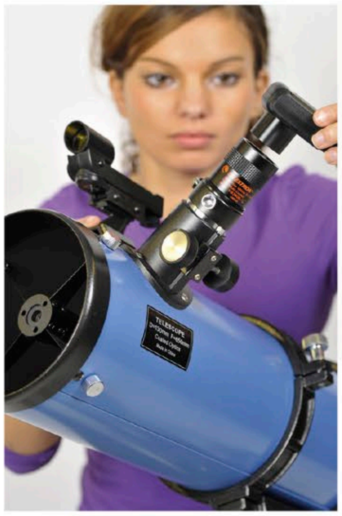
A telescope is, in effect, a very long telephoto lens, and adapters are available to attach virtually any single-lens reflex (SLR) camera to a telescope, thereby enabling the image produced by the telescope to be recorded. However, the maximum exposure time is often limited by the accuracy of the telescope's drive, which may not be precise enough to prevent star trailing. This problem can be overcome by using many short "sub-exposures" and adding them together with image-processing software to give the equivalent of a single long exposure. The telescope also needs to be kept steady, so a remote release should be used or, if possible, the camera should be operated remotely from a computer.

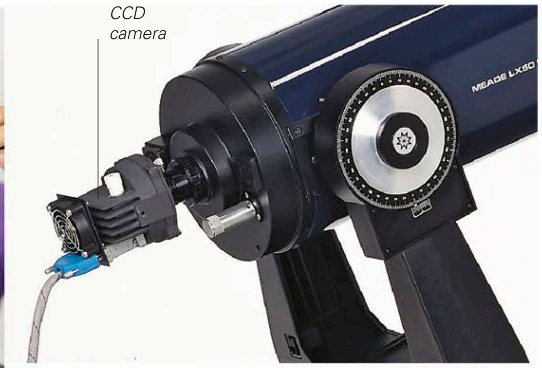
PRIME-FOCUS IMAGE OF THE DUMBBELL NEBULA
Prime-focus imaging is ideal for galaxies and small objects such as planetary nebulae—the Dumbbell Nebula shown here, for example. Exposure times of many minutes are needed for such images. To overcome any drive errors, the technique of sub-exposures (see above) can be used or a device called an autoguider can be fitted to the telescope to monitor the drive rate and make small corrections automatically.

WEBCAMS AND CCD IMAGING

Digital SLR cameras can produce good astronomical images but many advanced astrophotographers use either webcam-based cameras for planetary imaging or CCD cameras for imaging faint objects that require very long exposures. Planetary imaging is often badly affected by atmospheric turbulence, which typically blurs the view so that it is sharp for only fractions of a second. Webcam-type cameras produce a video stream, taking thousands of images a minute. These images can then be processed by dedicated software that selects and stacks together the best images. For imaging faint objects that require exposures of several hours, cooled CCD cameras produce less electronic noise—and therefore better images—than digital SLRs.

WEBCAM SET-UP
A webcam can be used on even small telescopes to image the planets. The webcam slots into the telescope in place of the eyepiece and connects to a computer with a cable. The webcam is then operated from the computer.



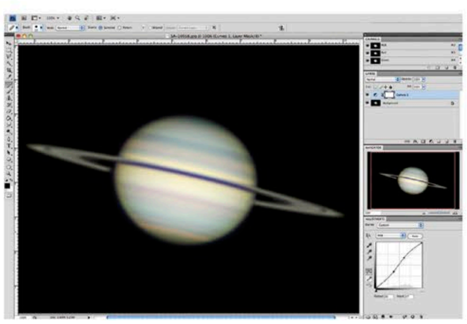


CCD IMAGING SET-UP
Like a webcam, a CCD camera replaces the telescope eyepiece and is connected to a computer. To quickly establish the focus when using a CCD camera, it helps first to focus using a telescope eyepiece with the same focus position as the camera.

IMAGE PROCESSING

Many images take far longer to process than the original observing time at the telescope, but there are various image-processing programs that can help. For example, software is available for automatically overlaying in register and stacking multiple exposures of the same object. Some cameras (notably CCD cameras) produce monochromatic images but can be used with color filters to produce a series of images that can be combined using stacking software to give a full-color final image. Software can also be used to enhance images by sharpening details, correcting the color balance, altering the brightness, and increasing the contrast. In addition, image-processing software can be used to change individual colors, a technique that is often utilized by professional astronomers to highlight specific features.

DIGITAL STACKING
The image (left) of NGC 1977, the Ghost Nebula in Orion, was made through an amateur 12.5 in (300 mm) telescope and is the result of combining four individual 90-minute exposures using dedicated image-stacking software.



COLOR CONTROL
This screenshot shows an image of Saturn in Photoshop, software that can be used to enhance or alter an image's features, such as its color. In this image, the brightest ring is composed of ice and needs to be altered to white to show a realistic view of the planet.